

EXACT REPRESENTATION-THEORETIC CHECKS FOR BINARY ICOSAHERAL CONSTRUCTIONS

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ABSTRACT. We report exact finite calculations for the binary icosahedral group $2I = \mathrm{SL}(2, 5)$, selected maps into exceptional Lie groups, and $\mathrm{SL}(2, \mathbb{Z}/70)$. We separate verified representation-theoretic facts from claims about compactifications and physical chirality. The character and Frobenius–Schur table, an eight-dimensional spinor-branching search and low-degree tensor invariants admit exact certificates. Several stronger exclusions in the original formulation do not follow: outer tensor products can carry mixed character fields, self-duality alone does not force a zero Dirac index, and selected examples do not classify all compactifications. The supplied scripts verify the explicitly stated finite scope, not a universal Standard-Model impossibility theorem.

1. SCOPE AND VERIFICATION CONVENTIONS

The computations use exact integer pairs $a + b\varphi$, where $\varphi^2 = \varphi + 1$, for the binary-icosahedral group. Quaternion coordinates are $(a + b\varphi)/2$. Finite class algebras and cyclotomic polynomials are handled by exact rational and algebraic arithmetic. A script failure is distinct from a disproved conjecture; uncomputed catalog or physics claims are recorded as unverified. The programs are executable certificates, not a proof-assistant formalization. This revision supersedes the mathematical claims of the historical preprint; the original source and PDF remain archived under their original names. Verification certificates apply only to the corrected scope stated here.

The internal space $S^3/2I \times T^2$ has real dimension five. Adding four-dimensional spacetime gives dimension nine. A ten-dimensional heterotic interpretation therefore requires additional explicitly specified structure; this product alone does not supply a six-dimensional internal space. No universal classification of such additional structure is attempted.

2. CHARACTERS AND ARITHMETIC

Proposition 2.1. *For a finite group G , all ordinary character values are algebraic integers in $\mathbb{Q}(\zeta_{\exp G})$. For $G = 2I$, the exponent is 60, and $\sin(\pi/14) \notin \mathbb{Q}(\zeta_{60})$. Moreover, $\sin(\pi/14)$ is not an ordinary character value of any finite group.*

Proof. The eigenvalues of g in a finite-dimensional representation are roots of unity whose orders divide $\exp G$, so their sum has the asserted properties. Set $z = e^{\pi i/7}$. Since $z^7 = -1$, $s = (z^3 - z^4)/2 = \cos(3\pi/7) = \sin(\pi/14)$. Exact reduction modulo $\Phi_{14}(z)$ gives $8s^3 - 4s^2 - 4s + 1 = 0$. The polynomial is irreducible over $\mathbb{Q}$; its possible rational roots $\pm 1, \pm 1/2, \pm 1/4, \pm 1/8$ are not roots. Thus $[\mathbb{Q}(s) : \mathbb{Q}] = 3$, which cannot divide $[\mathbb{Q}(\zeta_{60}) : \mathbb{Q}] = \phi(60) = 16$. The monic minimal polynomial $x^3 - \frac{1}{2}x^2 - \frac{1}{2}x + \frac{1}{8}$ is not in $\mathbb{Z}[x]$, so s is not an algebraic integer. This proves the literal-character assertion, not a statement about normalized observables or mixing angles. $\square$

The exact character field of $2I$ is $\mathbb{Q}(\sqrt{5})$; the exponent field above is an upper bound. In particular, arithmetic must be applied to the specific data and permitted operations, not simply to a level label.

Proposition 2.2. *The nine irreducible characters of $2I$ have the following dimensions and Frobenius–Schur indicators. Precisely the quaternionic irreducibles are faithful.*

Label	1	2_a	2_b	3_a	3_b	4_R	4_H	5	6
Dimension	1	2	2	3	3	4	4	5	6
Indicator	1	-1	-1	1	1	1	-1	1	-1

Proof. The 120 exact unit quaternions give nine conjugacy classes, with sizes

$$1, 1, 12, 12, 12, 12, 20, 20, 30.$$

Starting with the defining doublet, $\chi_{\text{Sym}^{n+1}} = \chi_2 \chi_{\text{Sym}^n} - \chi_{\text{Sym}^{n-1}}$ supplies the symmetric powers. The second doublet and triplet are their Galois conjugates; $4_R = 2_a \otimes 2_b$. These nine actual characters have exact inner products δ_{ij} and dimension-square sum 120. Evaluating $|G|^{-1} \sum_g \chi(g^2)$ gives the displayed indicators. Kernels are checked directly using $\chi(g) = \chi(1)$, and $\sum_\chi \nu(\chi) \chi(1) = 2 = \#\{g : g^2 = 1\}$. $\square$

Remark 2.3 (What self-duality does not imply). *This table does not prove that every equivariant Dirac index vanishes. An equivariant index is $[\ker D^+] - [\ker D^-] \in R(G)$, and a nonzero virtual representation can be self-dual. An antilinear map commuting with the chirality grading preserves, rather than exchanges, the two graded spaces. Vanishing requires additional structure pairing opposite-chirality kernels, compatible with the complete geometry and bundle twists. These conditions are not established by a character table alone.*

3. EIGHT-DIMENSIONAL SPINORS AND INVARIANTS

The real irreducible pieces have dimensions d for real type and $2d$ for quaternionic type. There are 25 multisets of total real dimension eight, 12 faithful. The programs construct lifts using $\text{Spin}(4) = \text{SU}(2) \times \text{SU}(2)$ and the lift of the realified $\text{Sym}^3 \mathbb{C}^2$ representation, not independent choices of angles on classes. For two four-dimensional blocks with half-spinors (A, B) and (C, D) ,

$$V = AB + CD, \quad S^+ = AC + BD, \quad S^- = AD + BC.$$

The realified 4_H has unordered half-spinor pair $\{3 \cdot 1 + 5, 2 \cdot 4_H\}$. Exact decomposition shows that it is the unique faithful real-eight module with three singlets in one half-spinor. This counts invariant spinors of the specified representations, not physical generations in an unspecified compactification.

For the defining doublet and triplet, $2_a \otimes 2_a = 1 + 3_a$, so this tensor product does contain a singlet. Using the harmonic decomposition $\text{Sym}^n(3_a) = V_n + V_{n-2} + \cdots$ gives

$$\dim(2_a \otimes 2_a \otimes \text{Sym}^n 3_a)^{2I} = 1, 1, 1, 1, 1, 2, 3, 3, 3, 4, 5, 6, 7 \quad (0 \leq n \leq 12).$$

The doublet Molien coefficients agree through degree 60 with $(1 + t^{30})/((1 - t^{12})(1 - t^{20}))$.

4. SELECTED CENTRALIZERS

Write V_j for the spin- j representation of $\text{SU}(2)$. For a principal $\text{SU}(2)$, the adjoint is the sum of V_j with j the Lie-algebra exponents, without added Cartan singlets. The programs recover these branchings from exact root and weight orbits and cross-check the E_8 branching against standard tables [7].

Specified restricted adjoint	Dimension of $2I$ invariants
Principal $\text{SU}(2)$ in E_8	0
Principal $\text{SU}(2)$ in E_6	0
Principal $\text{SU}(2)$ in $E_6 \subset E_8$	14
Regular $\text{SU}(2)$ in $\text{SU}(6) \times \text{SU}(2) \subset E_6 \subset E_8$	133
Regular $\text{SU}(2)$ in $E_7 \times \text{SU}(2) \subset E_8$	133

For the $E_6 \times A_2$ chain, one must retain the mixed terms:

$$248 = (78, 1) + (1, 8) + (27, 3) + (\overline{27}, \overline{3}).$$

The principal restriction of 27 is $V_0 + V_4 + V_8$, with one $2I$ invariant; thus the full invariant dimension is $0 + 8 + 6 = 14$. For the regular $\text{SU}(2)$, the corresponding invariant dimensions in

78 and 27 are 35 and 15, giving $35 + 8 + 6 \cdot 15 = 133$. The principal maps in E_6 and E_8 kill the central involution of $2I$; their finite image is A_5 , not an injectively embedded $2I$.

For the particular block embedding $\mathbb{R}^{16} = (\text{Sym}^3 \mathbb{C}^2)_{\mathbb{R}} \oplus \mathbb{R}^8$, the fixed dimensions in the D_8 adjoint and half-spinor are respectively 31 and 24. There is an additional identification argument, rather than a dimension-only guess: simultaneous triality on the two D_4 blocks preserves the 240 roots of E_8 and takes the torus weights $(3, 3, 1, 1, 0, 0, 0, 0)$ to $(4, 2, 0, 0, 0, 0, 0, 0)$. The vector then splits as $V_2 \oplus \mathbb{R}^{11}$, exhibiting a commuting $\mathfrak{so}(11)$ subalgebra. Under $\text{Spin}(5) \times \text{Spin}(11)$,

$$248 = (10, 1) + (1, 55) + (5, 11) + (4, 32),$$

whose first-factor restrictions are $V_1 + V_3$, V_0 , V_2 and $V_{3/2}$. The nontrivial summands have no $2I$ invariants. Thus the exhibited $\mathfrak{so}(11)$ exhausts the 55 fixed dimensions: the fixed Lie algebra is B_5 , of rank five, not $F_4 \oplus \mathfrak{u}(1)^3$. The global centralizer group and its components are not identified here.

The standard branching $248 = (52, 1) + (1, 14) + (26, 7)$ under $F_4 \times G_2$ contains no trivial summand, hence its Lie-algebra centralizer is zero [7]. Consequently no nontrivial *connected* holonomy subgroup can commute with this entire pair. A claim about the full group centralizer, disconnected holonomy, or an asymmetric orbifold requires additional arguments. $F_4 \times G_2$ is a special, non-regular subalgebra here; it is not a symmetric pair or a Dynkin-diagram folding of E_8 .

5. CONGRUENCE GROUPS AND COUNTEREXAMPLES TO BROAD EXCLUSIONS

CRT gives $\text{SL}(2, \mathbb{Z}/70) \cong S_3 \times 2I \times \text{SL}(2, 7)$. The group has order 241920, exponent 840, and 297 irreducibles; their Frobenius–Schur counts are 96 real, 93 quaternionic and 108 complex. There are eight three-dimensional irreducibles, none nontrivial on both the $2I$ and $\text{SL}(2, 7)$ factors. This limited dimension statement does not exclude interactions between fields in different representations.

Exact class-algebra diagonalization gives $\mathbb{Q}(\sqrt{2}, \sqrt{-7})$ as the character field of $\text{SL}(2, 7)$, agreeing with the independent table [1]. The product character field is

$$\mathbb{Q}(\sqrt{5}, \sqrt{2}, \sqrt{-7}), \quad [\mathbb{Q}(\sqrt{5}, \sqrt{2}, \sqrt{-7}) : \mathbb{Q}] = 8.$$

Neither $\sin(\pi/14)$ nor twice that number lies in this field. The former seven-element split-torus assignment to principal-series characters was incorrect.

This exclusion for one group must not be generalized to all direct products. In $2I \times D_{14}$, where D_{14} has order 14, the outer tensor product of the defining doublet with the dihedral doublet is irreducible and has a character value $\varphi(\zeta_7 + \zeta_7^{-1})$ of degree six. An exact resultant and irreducibility check certify this. Factorization of an ordinary character as a product does not make its representation reducible or prevent a compositum of character fields.

For $N \geq 3$, let $\mu = |\text{SL}(2, \mathbb{Z}/N)|$. The projective index is $\mu/2$, so $X(N)$ has $c = \mu/(2N)$ cusps and genus $g = 1 + \mu/24 - c/2$ [8]. At level 70 this gives $c = 1728$, $g = 9217$ and $\dim M_2 = 10944$. The weight-two newspace dimensions for $\Gamma_0(14)$, $\Gamma_0(35)$ and $\Gamma_0(70)$ are respectively 1, 3 and 1. Dimension is not the number of Galois orbits: level 35 has two orbits, one with coefficient field $\mathbb{Q}(\sqrt{17})$ [6, 4]. Modular-form coefficient fields must not be confused with finite-group character fields. At level 70, the label 70.2.a.a identifies the unique rational newform orbit, not an individual elliptic curve. LMFDB associates it with the conductor-70 elliptic-curve isogeny class 70.a (Cremona 70a) [5, 3]. This database identification supplies no numerical flavor fit or physical exclusion.

A zero Schur multiplier alone does not exclude central extensions with arbitrary finite coefficients when the abelianization is nonzero. The order-12 group

$$\{(g, t) \in S_3 \times \mathbb{Z}/4 : \epsilon(g) = t \bmod 2\}$$

with $\epsilon(g) \in \mathbb{Z}/2$ the permutation parity has central kernel $\mathbb{Z}/2$ and does not split over S_3 : each lift of a transposition has order four. Likewise, $\Delta(7350) = \Delta(150) \times_{S_3} \Delta(294)$, not their direct product.

For a twisted center $\mathcal{Z}(\text{Vec}_G^\omega)$, the untwisted exponent is not in general the twist-order bound. Already $G = \mathbb{Z}/2$ with $\omega(a, b, c) = (-1)^{abc}$ gives projective values $\pm i$ and twist order four. The group-theoretical Frobenius–Schur exponent instead divides $(\exp G)^2$ [2, Theorem 9.2]; for $2I$ the resulting bound is 3600. This concerns canonical categorical data, not arbitrary morphisms: in a complex-linear category the trace of λid_1 is λ . Finally, the second homotopy group of a 2-group is abelian; a proposed nonabelian $\pi_2 = \text{SL}(2, 7)$ is not such a homotopy datum. A crossed module must instead specify its boundary map, abelian kernel, action and Postnikov class.

6. TOPOLOGICAL ARITHMETIC AND LIMITS

For a verified free finite action, $\chi(X/G) = \chi(X)/|G|$. Integrality is necessary, not sufficient to construct the action. Singular quotients require fixed-point data rather than this free-cover formula. The boundary of the 600-cell triangulates S^3 and has Euler characteristic zero; the filled four-ball has Euler characteristic one. A covering deck action is an action on the covering space, not a newly established free action on the quotient.

The relation $|\chi|/2$ counts the magnitude of a net generation index only under the relevant Calabi–Yau threefold standard-embedding assumptions. For example, a *specified free* $\mathbb{Z}/2$ quotient of a $\chi = -12$ parent would have $\chi = -6$ and standard-embedding net count three. This arithmetic neither constructs nor excludes an asymmetric CHL model. No exhaustive Calabi–Yau action classification or numerical flavor-fit scan is supplied. Unsupported catalog exclusions and phenomenological fit estimates are therefore not promoted to propositions.

7. CONCLUSION

The exact finite calculations retain substantial representation-theoretic content while narrowing the conclusions. The literal-character arithmetic obstruction, the $2I$ Frobenius–Schur table, the specified spinor branchings, and the selected centralizer and congruence-group calculations are independently testable. They do not by themselves prove a universal chirality obstruction or rule out all finite-symmetry compactifications. A stronger claim must state its geometric, categorical and physical inputs and provide the corresponding proof or reproducible computation.

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